Supplementary material

Structural variation is invisible to SNP-distance dereplication: the reference strain of Yersinia pestis merges with its own attenuated derivative.

All tables below are generated from the archived derived tables by `scripts/make_tables.py` and typeset by `scripts/build_supplementary.py`. Each table carries the file it comes from. Supplementary Note 1 and Supplementary Note 2 are part of the manuscript text.

Table S1. Complete software version manifest, including the reason for each pinned dependency.

tool	version	notes
snippy	4.6.0	--ctgs --mincov 10 --minfrac 0.9
bakta	1.9.4	DB 5.1 full, MD5 f8823533b789dd315025fdcc46f1a8c1
pyrodigal	3.7.1	-
pyhmmer	0.10.15	pinned
AMRFinderPlus	4.2.7	DB 2026-08-07.1
NCBI Datasets CLI	18.38.0	taxon 632
SciPy	1.18.1	statistics
R	4.5.3	figure rendering: ggplot2 4.0.3, patchwork 1.3.2, ragg 1.5.2, ggrepel 0.9.8
R	4.4.3	remaining R analysis
MUMmer4/dnadiff	-	AlignedBases
CheckM	v1.2.4 (reported by NCBI)	completeness, contamination

source: pipeline/config.yaml; env/ENV_VERSIONS.txt; env/R_SESSION_INFO.txt

Table S2b. All genome assemblies analysed (accessions), in the order used by the pipeline.

accession	accession	accession	accession
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['GCF_000834495.1\r']	['GCA_030260845.1\r']	['GCA_046662065.1\r']	['GCA_046679285.1\r']
['GCF_000834775.1\r']	['GCA_030260855.1\r']	['GCA_046662085.1\r']	['GCA_046679305.1\r']
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['GCA_030260555.1\r']	['GCA_046661785.1\r']	['GCA_046679005.1\r']	['GCF_056640855.1\r']
['GCA_030260565.1\r']	['GCA_046661805.1\r']	['GCA_046679025.1\r']	['GCF_056640875.1\r']
['GCA_030260575.1\r']	['GCA_046661825.1\r']	['GCA_046679045.1\r']	['GCA_016102665.1\r']
['GCA_030260635.1\r']	['GCA_046661835.1\r']	['GCA_046679065.1\r']	['GCA_056640835.1\r']
['GCA_030260645.1\r']	['GCA_046661865.1\r']	['GCA_046679085.1\r']	['GCF_009295985.1\r']
['GCA_030260655.1\r']	['GCA_046661885.1\r']	['GCA_046679105.1\r']	['GCF_000986995.1\r']
['GCA_030260665.1\r']	['GCA_046661905.1\r']	['GCA_046679125.1\r']	['GCF_000022845.1\r']
['GCA_030260675.1\r']	['GCA_046661925.1\r']	['GCA_046679145.1\r']	
['GCA_030260735.1\r']	['GCA_046661945.1\r']	['GCA_046679165.1\r']	
['GCA_030260745.1\r']	['GCA_046661965.1\r']	['GCA_046679185.1\r']	

source: results/qc/tierB_candidates.tsv

Table S2a. The 161 distinct >=10 kb uncalled coordinates and how many genomes share each (5 kb-binned start).

replicon	start 5kb bin	length bp	n genomes
NC_003143	795000	15000	1127
NC_003134	0	20000	581
NC_003143	1310000	10000	505
NC_003143	2135000	105000	476
NC_003143	2370000	35000	173
NC_003143	2365000	25000	117
NC_003143	2745000	40000	89
NC_003143	615000	80000	89
NC_003143	2220000	15000	54
NC_003143	1105000	40000	53
NC_003143	810000	50000	50
NC_003143	4520000	45000	25
NC_003143	1110000	30000	21

replicon	start 5kb bin	length bp	n genomes
NC_003143	2190000	45000	17
NC_003143	1105000	30000	15
NC_003143	4135000	10000	15
NC_003143	2885000	30000	14
NC_003143	2555000	10000	11
NC_003143	3115000	10000	9
NC_003143	4020000	10000	6
NC_003143	2190000	50000	6
NC_003143	2200000	35000	5
NC_003143	615000	85000	5
NC_003143	1320000	40000	5
NC_003143	2365000	15000	5
NC_003143	3115000	15000	5
NC_003143	905000	65000	4
NC_003143	675000	10000	4
NC_003143	2220000	20000	4
NC_003143	2215000	25000	3
NC_003143	2605000	20000	3
NC_003143	2210000	30000	3
NC_003143	2135000	15000	3
NC_003143	2135000	20000	3
NC_003143	3050000	15000	3
NC_003143	2135000	35000	3
NC_003143	2210000	25000	3
NC_003143	2205000	35000	3
NC_003143	2200000	40000	3
NC_003143	4135000	15000	3
NC_003143	2135000	45000	3
NC_003143	795000	65000	3
NC_003143	2055000	15000	3
NC_003143	2545000	15000	3
NC_003143	4000000	45000	3
NC_003143	2175000	65000	3
NC_003143	660000	25000	3
NC_003143	2135000	10000	2
NC_003143	2600000	25000	2
NC_003143	2175000	60000	2
NC_003143	2190000	20000	2
NC_003143	2195000	45000	2
NC_003143	680000	15000	2
NC_003143	2170000	70000	2
NC_003143	2140000	25000	2
NC_003143	2185000	55000	2
NC_003143	2190000	25000	2
NC_003143	2215000	20000	2
NC_003143	2195000	10000	2
NC_003143	2135000	65000	2
NC_003143	2180000	20000	2
NC_003143	2135000	50000	2
NC_003143	2155000	20000	2
NC_003143	2180000	15000	2
NC_003143	2205000	15000	2
NC_003143	2175000	20000	2
NC_003143	2135000	55000	2
NC_003143	4015000	10000	2
NC_003143	2180000	60000	2
NC_003143	1230000	15000	2
NC_003143	3275000	10000	2
NC_003143	2175000	25000	1
NC_003143	2200000	10000	1
NC_003143	3115000	25000	1
NC_003143	650000	35000	1
NC_003143	2135000	75000	1
NC_003143	2135000	60000	1
NC_003143	1990000	25000	1
NC_003143	2150000	15000	1
NC_003143	2160000	75000	1
NC_003143	4530000	35000	1
NC_003143	845000	15000	1
NC_003143	2970000	15000	1
NC_003143	2170000	20000	1
NC_003143	635000	55000	1
NC_003143	680000	10000	1
NC_003143	2365000	40000	1
NC_003143	280000	10000	1

replicon	start 5kb bin	length bp	n genomes
NC_003143	2125000	135000	1
NC_003134	0	15000	1
NC_003143	2135000	70000	1
NC_003143	2155000	10000	1
NC_003143	660000	15000	1
NC_003143	2135000	80000	1
NC_003143	2145000	45000	1
NC_003143	2155000	30000	1
NC_003143	2135000	40000	1
NC_003143	2140000	70000	1
NC_003143	2135000	30000	1
NC_003143	2165000	15000	1
NC_003143	2180000	30000	1
NC_003143	655000	35000	1
NC_003143	2760000	25000	1
NC_003143	2155000	80000	1
NC_003143	2150000	45000	1
NC_003143	2185000	35000	1
NC_003143	2215000	15000	1
NC_003134	5000	15000	1
NC_003143	2175000	10000	1
NC_003143	2405000	10000	1
NC_003143	2150000	35000	1
NC_003143	2185000	15000	1
NC_003143	2160000	35000	1
NC_003143	2205000	10000	1
NC_003143	2140000	95000	1
NC_003143	2135000	25000	1
NC_003143	2160000	40000	1
NC_003143	2200000	15000	1
NC_003143	2210000	20000	1
NC_003143	1440000	65000	1
NC_003143	2190000	40000	1
NC_003143	2170000	65000	1
NC_003143	2190000	10000	1
NC_003143	2535000	10000	1
NC_003143	1235000	10000	1
NC_003143	1110000	40000	1
NC_003143	4000000	35000	1
NC_003134	0	40000	1
NC_003143	1245000	65000	1
NC_003143	3115000	35000	1
NC_003131	50000	15000	1
NC_003143	1980000	60000	1
NC_003143	1780000	30000	1
NC_003143	2140000	15000	1
NC_003143	2160000	15000	1
NC_003131	45000	20000	1
NC_003143	3660000	10000	1
NC_003143	3715000	30000	1
NC_003143	2135000	135000	1
NC_003143	2205000	25000	1
NC_003143	660000	30000	1
NC_003143	2190000	65000	1
NC_003143	2125000	130000	1
NC_003143	2190000	15000	1
NC_003143	2745000	80000	1
NC_003143	675000	15000	1
NC_003143	2785000	20000	1
NC_003143	2185000	50000	1
NC_003143	1065000	15000	1
NC_003143	1655000	15000	1
NC_003143	4210000	15000	1
NC_003143	2550000	10000	1
NC_003143	2145000	10000	1
NC_003143	4510000	10000	1
NC_003143	2145000	15000	1
NC_003143	885000	15000	1
NC_003143	2665000	10000	1
NC_003143	685000	15000	1
NC_003143	2760000	10000	1
NC_003143	1895000	30000	1
NC_003143	3730000	10000	1

source: results/qc/uncalled_blocks_all.tsv

Table S3a. Threshold sweep on Tier A: retained genomes, largest cluster diameter and size, and same-strain pairs merged.

t	retained	max diameter	max size	same strain pairs merged
0	249	0	3	0
3	238	3	4	1
5	226	5	4	2
10	210	10	5	3
15	189	15	8	4
20	175	20	12	5
25	167	25	13	5
30	154	30	14	5
50	93	50	18	5
75	49	75	24	5
100	37	98	37	5

source: results/figures/fig4d_sweep.csv

Table S3b. Threshold sweep on Tier B, including the two off-grid integers (t = 18, 19) that bracket the arithmetic floor.

t	retained	max size	grid point
0	2704	16	grid
3	1653	39	grid
5	1239	50	grid
10	750	156	grid
15	536	319	grid
18	453	397	off-grid
19	429	464	off-grid
20	410	464	grid
25	352	464	grid
30	309	464	grid
50	190	584	grid
75	120	1108	grid
100	95	1108	grid

source: results/dedup/snp_distance_raw_tierB.npy (complete linkage)

Table S4. Plasmid three-state reclassification of the 258 closed genomes from each assembly's own submitted sequence inventory.

accession	n plasmid sequences	pCD1	pPCP1	pMT1
GCA_000006645.1	1	LOST	LOST	CARRIED
GCA_053570175.1	3	CARRIED	CARRIED	CARRIED
GCA_053570195.1	3	CARRIED	CARRIED	CARRIED
GCA_053570215.1	3	CARRIED	CARRIED	CARRIED
GCA_053570235.1	3	CARRIED	CARRIED	CARRIED
GCA_053570255.1	3	CARRIED	CARRIED	CARRIED
GCA_053570275.1	3	CARRIED	CARRIED	CARRIED
GCA_053570295.1	4	CARRIED	CARRIED	CARRIED
GCA_053570315.1	3	CARRIED	CARRIED	CARRIED
GCA_053570335.1	3	CARRIED	CARRIED	CARRIED
GCA_053570355.1	3	CARRIED	CARRIED	CARRIED
GCA_053570365.1	3	CARRIED	CARRIED	CARRIED
GCA_053570395.1	3	CARRIED	CARRIED	CARRIED
GCA_053570405.1	3	CARRIED	CARRIED	CARRIED
GCA_053570435.1	3	CARRIED	CARRIED	CARRIED
GCA_053570445.1	3	CARRIED	CARRIED	CARRIED
GCA_062019011.1	3	CARRIED	CARRIED	CARRIED
GCF_000006645.1	1	LOST	LOST	CARRIED
GCF_000007885.1	4	CARRIED	CARRIED	CARRIED
GCF_000009065.1	3	CARRIED	CARRIED	CARRIED
GCF_000013805.1	2	LOST	CARRIED	CARRIED
GCF_000013825.1	3	CARRIED	CARRIED	CARRIED
GCF_000016445.1	2	CARRIED	LOST	LOST
GCF_000018805.1	2	CARRIED	LOST	CARRIED
GCF_000022805.1	3	CARRIED	CARRIED	CARRIED
GCF_000022825.1	3	CARRIED	CARRIED	CARRIED
GCF_000022845.1	3	CARRIED	CARRIED	CARRIED
GCF_000186725.1	3	CARRIED	CARRIED	CARRIED
GCF_000222975.1	2	LOST	CARRIED	CARRIED
GCF_000590535.2	0	UNRESOLVED	UNRESOLVED	UNRESOLVED
GCF_000834235.1	3	CARRIED	CARRIED	CARRIED
GCF_000834275.1	3	CARRIED	CARRIED	CARRIED
GCF_000834315.1	2	CARRIED	LOST	LOST
GCF_000834335.1	3	CARRIED	CARRIED	CARRIED
GCF_000834495.1	3	CARRIED	CARRIED	CARRIED

accession	n plasmid sequences	pCD1	pPCP1	pMT1
GCF_000834755.1	2	LOST	CARRIED	CARRIED
GCF_000834775.1	2	CARRIED	LOST	LOST
GCF_000834825.1	3	CARRIED	CARRIED	CARRIED
GCF_000834845.1	3	CARRIED	CARRIED	CARRIED
GCF_000834885.1	3	CARRIED	CARRIED	CARRIED
GCF_000834905.1	4	CARRIED	CARRIED	LOST
GCF_000834925.1	3	CARRIED	CARRIED	CARRIED
GCF_000834985.1	2	CARRIED	LOST	LOST
GCF_000970105.1	3	CARRIED	CARRIED	CARRIED
GCF_001188675.1	2	LOST	CARRIED	LOST
GCF_001188695.1	2	CARRIED	LOST	LOST
GCF_001188715.1	2	CARRIED	LOST	LOST
GCF_001188735.1	3	CARRIED	CARRIED	CARRIED
GCF_001188755.1	2	CARRIED	LOST	CARRIED
GCF_001188775.1	2	CARRIED	LOST	CARRIED
GCF_001188795.1	2	CARRIED	LOST	CARRIED
GCF_001188815.1	3	CARRIED	CARRIED	CARRIED
GCF_001188935.1	2	CARRIED	LOST	LOST
GCF_001293415.1	2	CARRIED	LOST	CARRIED
GCF_001693595.1	3	CARRIED	CARRIED	CARRIED
GCF_002005285.1	3	CARRIED	CARRIED	CARRIED
GCF_003798205.1	4	CARRIED	CARRIED	CARRIED
GCF_003798225.1	3	CARRIED	LOST	LOST
GCF_003798345.1	2	CARRIED	LOST	CARRIED
GCF_009295925.1	3	CARRIED	CARRIED	CARRIED
GCF_009295945.1	4	CARRIED	CARRIED	CARRIED
GCF_009295965.1	3	CARRIED	CARRIED	CARRIED
GCF_009295985.1	4	CARRIED	CARRIED	CARRIED
GCF_009296005.1	2	LOST	CARRIED	CARRIED
GCF_009363195.1	4	CARRIED	CARRIED	CARRIED
GCF_015159615.2	0	UNRESOLVED	UNRESOLVED	UNRESOLVED
GCF_015190655.1	4	CARRIED	CARRIED	CARRIED
GCF_015336085.1	0	UNRESOLVED	UNRESOLVED	UNRESOLVED
GCF_015336265.1	0	UNRESOLVED	UNRESOLVED	UNRESOLVED
GCF_015336465.1	0	UNRESOLVED	UNRESOLVED	UNRESOLVED
GCF_015336695.1	0	UNRESOLVED	UNRESOLVED	UNRESOLVED
GCF_015336865.1	0	UNRESOLVED	UNRESOLVED	UNRESOLVED
GCF_015337085.2	0	UNRESOLVED	UNRESOLVED	UNRESOLVED
GCF_015337285.1	0	UNRESOLVED	UNRESOLVED	UNRESOLVED
GCF_015337445.1	0	UNRESOLVED	UNRESOLVED	UNRESOLVED
GCF_015337645.1	0	UNRESOLVED	UNRESOLVED	UNRESOLVED
GCF_015337825.2	0	UNRESOLVED	UNRESOLVED	UNRESOLVED
GCF_015338045.2	0	UNRESOLVED	UNRESOLVED	UNRESOLVED
GCF_015338205.1	0	UNRESOLVED	UNRESOLVED	UNRESOLVED
GCF_024498375.1	2	CARRIED	LOST	LOST
GCF_024498395.1	3	CARRIED	LOST	CARRIED
GCF_024498415.1	3	CARRIED	LOST	LOST
GCF_024758685.1	3	CARRIED	LOST	LOST
GCF_039640175.1	3	CARRIED	CARRIED	CARRIED
GCF_039640395.1	3	CARRIED	CARRIED	CARRIED
GCF_039640405.1	3	CARRIED	CARRIED	CARRIED
GCF_039640415.1	3	CARRIED	CARRIED	CARRIED
GCF_039640425.1	3	CARRIED	CARRIED	CARRIED
GCF_039640435.1	3	CARRIED	CARRIED	CARRIED
GCF_039640445.1	3	CARRIED	CARRIED	CARRIED
GCF_039640455.1	3	CARRIED	CARRIED	CARRIED
GCF_039640465.1	3	CARRIED	CARRIED	CARRIED
GCF_039640475.1	3	CARRIED	CARRIED	CARRIED
GCF_039640485.1	3	CARRIED	CARRIED	CARRIED
GCF_039703215.1	3	CARRIED	LOST	CARRIED
GCF_039703785.1	3	CARRIED	CARRIED	CARRIED
GCF_041409745.1	2	CARRIED	LOST	CARRIED
GCF_041410195.1	3	CARRIED	CARRIED	CARRIED
GCF_041937415.1	2	LOST	CARRIED	CARRIED
GCF_046895365.1	3	CARRIED	CARRIED	CARRIED
GCF_046895375.1	3	CARRIED	CARRIED	CARRIED
GCF_046895395.1	4	CARRIED	CARRIED	CARRIED
GCF_046895405.1	2	CARRIED	LOST	CARRIED
GCF_046895415.1	2	CARRIED	LOST	CARRIED
GCF_046895425.1	3	LOST	CARRIED	CARRIED
GCF_046895435.1	3	CARRIED	CARRIED	CARRIED
GCF_046895445.1	3	CARRIED	CARRIED	CARRIED
GCF_046895455.1	3	CARRIED	CARRIED	CARRIED
GCF_046895465.1	3	CARRIED	CARRIED	CARRIED
GCF_046895475.1	3	CARRIED	CARRIED	CARRIED

accession	n plasmid sequences	pCD1	pPCP1	pMT1
GCF_046895485.1	3	CARRIED	CARRIED	CARRIED
GCF_046895495.1	3	CARRIED	CARRIED	CARRIED
GCF_046895505.1	3	CARRIED	CARRIED	CARRIED
GCF_046895515.1	3	CARRIED	CARRIED	CARRIED
GCF_046895525.1	3	CARRIED	CARRIED	CARRIED
GCF_046895535.1	3	CARRIED	CARRIED	CARRIED
GCF_046895545.1	4	CARRIED	CARRIED	CARRIED
GCF_046895555.1	4	CARRIED	CARRIED	CARRIED
GCF_046895565.1	4	CARRIED	CARRIED	CARRIED
GCF_046895575.1	3	CARRIED	CARRIED	CARRIED
GCF_046895585.1	3	CARRIED	CARRIED	CARRIED
GCF_046895595.1	3	CARRIED	CARRIED	CARRIED
GCF_046895605.1	3	CARRIED	CARRIED	CARRIED
GCF_046895615.1	3	CARRIED	CARRIED	CARRIED
GCF_046895625.1	3	CARRIED	CARRIED	CARRIED
GCF_046895635.1	5	CARRIED	CARRIED	CARRIED
GCF_046895645.1	3	CARRIED	CARRIED	CARRIED
GCF_046895655.1	3	CARRIED	CARRIED	CARRIED
GCF_046895665.1	4	CARRIED	CARRIED	CARRIED
GCF_046895675.1	4	CARRIED	CARRIED	CARRIED
GCF_046895685.1	4	CARRIED	CARRIED	CARRIED
GCF_046895695.1	4	CARRIED	CARRIED	CARRIED
GCF_046895705.1	3	CARRIED	CARRIED	CARRIED
GCF_046895715.1	3	CARRIED	CARRIED	CARRIED
GCF_046895725.1	3	CARRIED	CARRIED	CARRIED
GCF_046895735.1	4	CARRIED	CARRIED	CARRIED
GCF_046895745.1	4	CARRIED	CARRIED	CARRIED
GCF_046895755.1	3	CARRIED	CARRIED	CARRIED
GCF_046895765.1	3	CARRIED	CARRIED	CARRIED
GCF_046895775.1	3	CARRIED	CARRIED	CARRIED
GCF_046895785.1	3	CARRIED	CARRIED	CARRIED
GCF_046895795.1	3	CARRIED	CARRIED	CARRIED
GCF_046895805.1	3	CARRIED	CARRIED	CARRIED
GCF_046895815.1	3	CARRIED	CARRIED	CARRIED
GCF_046895825.1	3	CARRIED	CARRIED	CARRIED
GCF_046895835.1	3	CARRIED	CARRIED	CARRIED
GCF_046895845.1	3	CARRIED	CARRIED	CARRIED
GCF_046895855.1	3	CARRIED	CARRIED	CARRIED
GCF_046895865.1	3	CARRIED	CARRIED	CARRIED
GCF_046895875.1	3	CARRIED	CARRIED	CARRIED
GCF_046895885.1	3	CARRIED	CARRIED	CARRIED
GCF_046895895.1	3	CARRIED	CARRIED	CARRIED
GCF_046895905.1	3	CARRIED	CARRIED	CARRIED
GCF_046895915.1	3	CARRIED	CARRIED	CARRIED
GCF_046895925.1	3	CARRIED	CARRIED	CARRIED
GCF_046895935.1	3	CARRIED	CARRIED	CARRIED
GCF_046895945.1	3	CARRIED	CARRIED	CARRIED
GCF_046895955.1	3	CARRIED	CARRIED	CARRIED
GCF_046895965.1	3	CARRIED	CARRIED	CARRIED
GCF_046895975.1	3	CARRIED	CARRIED	CARRIED
GCF_046895985.1	3	CARRIED	CARRIED	CARRIED
GCF_052373935.1	3	CARRIED	CARRIED	CARRIED
GCF_052373945.1	3	CARRIED	CARRIED	CARRIED
GCF_052373955.1	3	CARRIED	CARRIED	CARRIED
GCF_052373995.1	4	CARRIED	CARRIED	CARRIED
GCF_052374015.1	3	CARRIED	CARRIED	CARRIED
GCF_052374055.1	3	CARRIED	CARRIED	CARRIED
GCF_052374075.1	3	CARRIED	CARRIED	CARRIED
GCF_052374095.1	3	CARRIED	CARRIED	CARRIED
GCF_052374115.1	3	CARRIED	CARRIED	CARRIED
GCF_052374135.1	3	CARRIED	CARRIED	CARRIED
GCF_052374155.1	3	CARRIED	CARRIED	CARRIED
GCF_052374175.1	4	CARRIED	CARRIED	CARRIED
GCF_052374195.1	3	CARRIED	CARRIED	CARRIED
GCF_052374215.1	3	CARRIED	CARRIED	CARRIED
GCF_052374235.1	3	CARRIED	CARRIED	CARRIED
GCF_052374255.1	3	CARRIED	CARRIED	CARRIED
GCF_052374275.1	3	CARRIED	CARRIED	CARRIED
GCF_052374295.1	3	CARRIED	CARRIED	CARRIED
GCF_052374315.1	3	CARRIED	CARRIED	CARRIED
GCF_052374335.1	3	CARRIED	CARRIED	CARRIED
GCF_052374355.1	3	CARRIED	CARRIED	CARRIED
GCF_052374375.1	3	CARRIED	CARRIED	CARRIED
GCF_052374395.1	3	CARRIED	CARRIED	CARRIED
GCF_052374415.1	3	CARRIED	CARRIED	CARRIED

accession	n plasmid sequences	pCD1	pPCP1	pMT1
GCF_052374445.1	3	CARRIED	CARRIED	CARRIED
GCF_052374465.1	3	CARRIED	CARRIED	CARRIED
GCF_052374495.1	3	CARRIED	CARRIED	CARRIED
GCF_052374515.1	3	CARRIED	CARRIED	CARRIED
GCF_052374535.1	3	CARRIED	CARRIED	CARRIED
GCF_052374555.1	3	CARRIED	CARRIED	CARRIED
GCF_052374575.1	3	CARRIED	CARRIED	CARRIED
GCF_052374595.1	3	CARRIED	CARRIED	CARRIED
GCF_052374615.1	3	CARRIED	CARRIED	CARRIED
GCF_052374635.1	3	CARRIED	CARRIED	CARRIED
GCF_052374655.1	3	CARRIED	CARRIED	CARRIED
GCF_052374675.1	3	CARRIED	CARRIED	CARRIED
GCF_052374695.1	3	CARRIED	CARRIED	CARRIED
GCF_052374715.1	3	CARRIED	CARRIED	CARRIED
GCF_052374735.1	3	CARRIED	CARRIED	CARRIED
GCF_052374755.1	3	CARRIED	CARRIED	CARRIED
GCF_052374775.1	3	CARRIED	CARRIED	CARRIED
GCF_052374795.1	3	CARRIED	CARRIED	CARRIED
GCF_052374815.1	3	CARRIED	CARRIED	CARRIED
GCF_052374855.1	3	CARRIED	CARRIED	CARRIED
GCF_052374875.1	3	CARRIED	CARRIED	CARRIED
GCF_052374895.1	3	CARRIED	CARRIED	CARRIED
GCF_052374915.1	3	CARRIED	CARRIED	CARRIED
GCF_052374935.1	3	CARRIED	CARRIED	CARRIED
GCF_052374955.1	3	CARRIED	CARRIED	CARRIED
GCF_052374975.1	3	CARRIED	CARRIED	CARRIED
GCF_052374995.1	3	CARRIED	CARRIED	CARRIED
GCF_052375015.1	3	CARRIED	CARRIED	CARRIED
GCF_052375035.1	3	CARRIED	CARRIED	CARRIED
GCF_052375045.1	3	CARRIED	CARRIED	CARRIED
GCF_052375075.1	3	CARRIED	CARRIED	CARRIED
GCF_052375095.1	3	CARRIED	CARRIED	CARRIED
GCF_052375105.1	3	CARRIED	CARRIED	CARRIED
GCF_052375135.1	3	CARRIED	CARRIED	CARRIED
GCF_052375155.1	3	CARRIED	CARRIED	CARRIED
GCF_052375175.1	3	CARRIED	CARRIED	CARRIED
GCF_052375195.1	3	CARRIED	CARRIED	CARRIED
GCF_052375235.1	3	CARRIED	CARRIED	CARRIED
GCF_052375255.1	3	CARRIED	CARRIED	CARRIED
GCF_052375265.1	3	CARRIED	CARRIED	CARRIED
GCF_052375295.1	3	CARRIED	CARRIED	CARRIED
GCF_052375315.1	3	CARRIED	CARRIED	CARRIED
GCF_052375335.1	3	CARRIED	CARRIED	CARRIED
GCF_052375345.1	3	CARRIED	CARRIED	CARRIED
GCF_052375375.1	3	CARRIED	CARRIED	CARRIED
GCF_052375395.1	3	CARRIED	CARRIED	CARRIED
GCF_052375415.1	3	CARRIED	CARRIED	CARRIED
GCF_052375435.1	3	CARRIED	CARRIED	CARRIED
GCF_052375455.1	3	CARRIED	CARRIED	CARRIED
GCF_052375475.1	3	CARRIED	CARRIED	CARRIED
GCF_052375495.1	3	CARRIED	CARRIED	CARRIED
GCF_052375515.1	3	CARRIED	CARRIED	CARRIED
GCF_052375535.1	3	CARRIED	CARRIED	CARRIED
GCF_052375555.1	3	CARRIED	CARRIED	CARRIED
GCF_052375615.1	3	CARRIED	CARRIED	CARRIED
GCF_052375635.1	3	CARRIED	CARRIED	CARRIED
GCF_052375655.1	3	CARRIED	CARRIED	CARRIED
GCF_052375675.1	3	CARRIED	CARRIED	CARRIED
GCF_052375695.1	3	CARRIED	CARRIED	CARRIED
GCF_052375715.1	3	CARRIED	CARRIED	CARRIED
GCF_052375735.1	3	CARRIED	CARRIED	CARRIED
GCF_052375755.1	3	CARRIED	CARRIED	CARRIED
GCF_052375775.1	3	CARRIED	CARRIED	CARRIED
GCF_052375795.1	3	CARRIED	CARRIED	CARRIED
GCF_052375815.1	3	CARRIED	CARRIED	CARRIED
GCF_052375835.1	3	CARRIED	CARRIED	CARRIED
GCF_052375915.1	3	CARRIED	CARRIED	CARRIED
GCF_052446015.1	3	CARRIED	CARRIED	CARRIED
GCF_053834845.1	3	CARRIED	CARRIED	CARRIED
GCF_053834855.1	3	CARRIED	CARRIED	CARRIED
GCF_053834865.1	3	CARRIED	CARRIED	CARRIED
GCF_053834875.1	3	CARRIED	CARRIED	CARRIED
GCF_061402485.1	2	CARRIED	LOST	CARRIED

source: results/qc/replicon_three_state_seqreport.tsv + assembly_metadata.tsv

Table S5. Per-genome classification of the baseline layer, with the S1/S2 decompositions.

accession	assembly level	laboratory named	replicons absent	big blocks 5kb	n big blocks	strain
GCA_000006645.1	Complete Genome	no	NC_003131;NC_003132	NC_003143:1310000:10000;NC_003143:795000:15000	2	KIM
GCA_053570175.1	Chromosome	no		NC_003143:1110000:30000	1	66_YP38_PB
GCA_053570195.1	Chromosome	no		NC_003143:1110000:30000	1	48_YP14_PAK
GCA_053570215.1	Chromosome	no		NC_003143:1110000:30000	1	37_YP35_TLH
GCA_053570235.1	Chromosome	no		NC_003143:1110000:30000	1	31_YP74_SZ
GCA_053570255.1	Chromosome	no		NC_003143:1110000:30000	1	31_YP37_SZ
GCA_053570275.1	Chromosome	no		NC_003143:1110000:30000	1	28_YP48_MKUM
GCA_053570295.1	Chromosome	no		NC_003143:1110000:30000	1	23_YP47_MANG
GCA_053570315.1	Chromosome	no		NC_003143:1110000:30000	1	23_YP48_MANG
GCA_053570335.1	Chromosome	no		NC_003143:1110000:30000	1	23_YP45_MANG
GCA_053570355.1	Chromosome	no		NC_003143:1110000:30000	1	21_YP14_NP
GCA_053570365.1	Chromosome	no		NC_003143:1110000:30000	1	20_YP79_UST
GCA_053570395.1	Chromosome	no		NC_003143:1110000:30000	1	20_YP78_UST
GCA_053570405.1	Chromosome	no		NC_003143:1110000:30000	1	19_YP08_PrU
GCA_053570435.1	Chromosome	no		NC_003143:1110000:30000;NC_003143:1235000:10000	2	15_YP93_VU
GCA_053570445.1	Chromosome	no		NC_003143:1110000:30000	1	16_YP27_VU
GCA_062019011.1	Complete Genome	no			0	JC25
GCF_000006645.1	Complete Genome	yes	NC_003131;NC_003132	NC_003143:1310000:10000;NC_003143:795000:15000	2	KIM10+
GCF_000007885.1	Complete Genome	no		NC_003143:2370000:35000	1	91001
GCF_000009065.1	Complete Genome	yes			0	CO92
GCF_000013805.1	Complete Genome	yes	NC_003131	NC_003143:2135000:105000;NC_003143:795000:15000	2	Nepal516
GCF_000013825.1	Complete Genome	yes		NC_003143:2365000:15000;NC_003143:2555000:10000	2	Antiqua
GCF_000016445.1	Complete Genome	yes	NC_003132	NC_003143:1105000:40000;NC_003143:795000:65000	2	Pestoides F
GCF_000018805.1	Complete Genome	yes		NC_003143:1105000:40000;NC_003143:1310000:10000;NC_003143:2055000:15000;NC_003143:2545000:15000;NC_003143:4000000:45000;NC_003143:810000:50000	6	Angola
GCF_000022805.1	Complete Genome	no			0	D106004
GCF_000022825.1	Complete Genome	no			0	D182038
GCF_000022845.1	Complete Genome	no		NC_003143:2135000:105000;NC_003143:4520000:45000	2	Z176003
GCF_000186725.1	Complete Genome	no		NC_003143:2135000:105000;NC_003143:795000:15000	2	Harbin 35
GCF_000222975.1	Complete Genome	yes	NC_003131	NC_003143:2135000:105000	1	A1122

accession	assembly level	laboratory named	replicons absent	big blocks 5kb	n big blocks	strain
GCF_000590535.2	Complete Genome	yes	NC_003131;NC_003132;NC_003134	NC_003143:1310000:10000;NC_003143:2135000:105000	2	EV NIIEG
GCF_000834235.1	Complete Genome	no			0	PBM19
GCF_000834275.1	Complete Genome	yes		NC_003143:2135000:105000;NC_003143:795000:15000	2	Harbin35
GCF_000834315.1	Complete Genome	yes	NC_003132	NC_003143:1105000:40000;NC_003143:795000:65000	2	Pestoides F
GCF_000834335.1	Complete Genome	no			0	Shasta
GCF_000834495.1	Complete Genome	no			0	El Dorado
GCF_000834755.1	Complete Genome	yes	NC_003131	NC_003143:2135000:105000	1	A1122
GCF_000834775.1	Complete Genome	no	NC_003132	NC_003134:0:40000	1	Dodson
GCF_000834825.1	Complete Genome	yes		NC_003143:2365000:15000;NC_003143:2555000:10000	2	Antiqua
GCF_000834845.1	Complete Genome	yes		NC_003143:1105000:40000;NC_003143:1310000:10000;NC_003143:2055000:15000;NC_003143:2545000:15000;NC_003143:4000000:45000;NC_003143:810000:50000	6	Angola
GCF_000834885.1	Complete Genome	no		NC_003143:2135000:105000;NC_003143:795000:15000	2	Nicholisk 41
GCF_000834905.1	Chromosome	yes	NC_003134		0	Java9
GCF_000834925.1	Complete Genome	no		NC_003143:2370000:35000	1	Pestoides B
GCF_000834985.1	Complete Genome	no	NC_003132	NC_003143:1105000:40000;NC_003143:810000:50000	2	Pestoides G
GCF_000970105.1	Chromosome	no		NC_003143:1310000:10000;NC_003143:795000:15000	2	KIM5
GCF_001188675.1	Chromosome	no		NC_003131:50000:15000;NC_003143:1110000:30000	2	790
GCF_001188695.1	Complete Genome	no	NC_003132	NC_003143:1105000:40000;NC_003143:810000:50000	2	1412
GCF_001188715.1	Complete Genome	no	NC_003132	NC_003143:1105000:40000;NC_003143:810000:50000	2	1522
GCF_001188735.1	Chromosome	no		NC_003143:1310000:10000;NC_003143:2135000:105000;NC_003143:795000:15000	3	1045
GCF_001188755.1	Complete Genome	no	NC_003132	NC_003143:1105000:40000;NC_003143:1980000:60000;NC_003143:810000:50000	3	8787
GCF_001188775.1	Complete Genome	no	NC_003132	NC_003143:1105000:40000;NC_003143:810000:50000	2	3770
GCF_001188795.1	Complete Genome	no	NC_003132	NC_003143:1105000:40000;NC_003143:810000:50000	2	3067

accession	assembly level	laboratory named	replicons absent	big blocks 5kb	n big blocks	strain
GCF_001188815.1	Complete Genome	no		NC_003143:1310000:10000;NC_003143:2135000:105000;NC_003143:795000:15000	3	2944
GCF_001188935.1	Complete Genome	no	NC_003132	NC_003143:1105000:40000;NC_003143:810000:50000	2	1413
GCF_001293415.1	Complete Genome	yes	NC_003132	NC_003143:2135000:105000	1	CO92
GCF_001693595.1	Complete Genome	no		NC_003143:2135000:105000	1	Cadman
GCF_002005285.1	Complete Genome	no			0	195/P
GCF_003798205.1	Complete Genome	no		NC_003143:795000:15000	1	FDAARGOS_603
GCF_003798225.1	Complete Genome	no		NC_003143:2365000:15000;NC_003143:2555000:10000	2	FDAARGOS_601
GCF_003798345.1	Complete Genome	no	NC_003132	NC_003143:2370000:35000	1	FDAARGOS_602
GCF_009295925.1	Complete Genome	no		NC_003143:1110000:30000	1	SCPM-O-B-6899 (231)
GCF_009295945.1	Complete Genome	no		NC_003143:2135000:105000	1	SCPM-O-DNA-18 (I-3113)
GCF_009295965.1	Complete Genome	no		NC_003143:795000:15000	1	SCPM-O-B-5935 (I-1996)
GCF_009295985.1	Complete Genome	no		NC_003143:2180000:60000;NC_003143:795000:15000	2	SCPM-O-B-6530
GCF_009296005.1	Complete Genome	yes	NC_003131	NC_003143:1310000:10000;NC_003143:2135000:105000;NC_003143:795000:15000	3	SCPM-O-B-6291 (C-25)
GCF_009363195.1	Complete Genome	no		NC_003143:1230000:15000;NC_003143:1310000:10000	2	SCPM-O-B-5942 (I-2638)
GCF_015159615.2	Complete Genome	no	NC_003131;NC_003132;NC_003134	NC_003143:1310000:10000;NC_003143:795000:15000	2	14D
GCF_015190655.1	Complete Genome	no		NC_003143:795000:15000	1	S19960127
GCF_015336085.1	Complete Genome	no	NC_003131;NC_003132;NC_003134	NC_003143:1310000:10000;NC_003143:795000:15000	2	C-781
GCF_015336265.1	Complete Genome	no	NC_003131;NC_003132;NC_003134	NC_003143:1310000:10000	1	M2029
GCF_015336465.1	Complete Genome	no	NC_003131;NC_003132;NC_003134	NC_003143:3050000:15000;NC_003143:3275000:10000;NC_003143:795000:15000	3	I-1252
GCF_015336695.1	Complete Genome	no	NC_003131;NC_003132;NC_003134	NC_003143:1310000:10000;NC_003143:795000:15000	2	M2086
GCF_015336865.1	Complete Genome	no	NC_003131;NC_003132;NC_003134	NC_003143:2370000:35000	1	M-1974
GCF_015337085.2	Complete Genome	yes	NC_003131;NC_003132;NC_003134	NC_003143:1310000:10000;NC_003143:2175000:65000;NC_003143:2745000:80000;NC_003143:795000:15000	4	C-792
GCF_015337285.1	Complete Genome	yes	NC_003131;NC_003132;NC_003134	NC_003143:2135000:105000;NC_003143:795000:15000	2	C-783
GCF_015337445.1	Complete Genome	no	NC_003131;NC_003132;NC_003134	NC_003143:1310000:10000;NC_003143:2135000:105000;NC_003143:795000:15000	3	KM 567

accession	assembly level	laboratory named	replicons absent	big blocks 5kb	n big blocks	strain
GCF_015337645.1	Complete Genome	yes	NC_003131;NC_003132;NC_003134	NC_003143:1310000:10000;NC_003143:795000:15000	2	M-1482
GCF_015337825.2	Complete Genome	no	NC_003131;NC_003132;NC_003134	NC_003143:1310000:10000;NC_003143:795000:15000	2	M-1770
GCF_015338045.2	Complete Genome	no	NC_003131;NC_003132;NC_003134	NC_003143:1230000:15000;NC_003143:1310000:10000	2	M2085
GCF_015338205.1	Complete Genome	yes	NC_003131;NC_003132;NC_003134	NC_003143:1105000:40000;NC_003143:2885000:30000;NC_003143:3115000:10000;NC_003143:810000:50000	4	C-830
GCF_024498375.1	Complete Genome	no			0	R
GCF_024498395.1	Complete Genome	no			0	94
GCF_024498415.1	Complete Genome	no		NC_003134:0:20000	1	20
GCF_024758685.1	Complete Genome	yes		NC_003143:1310000:10000;NC_003143:2135000:105000	2	EV76-CN
GCF_039640175.1	Complete Genome	no			0	133
GCF_039640395.1	Complete Genome	no			0	136
GCF_039640405.1	Complete Genome	no		NC_003143:1310000:10000;NC_003143:2745000:40000;NC_003143:615000:80000	3	A1694
GCF_039640415.1	Complete Genome	no			0	91_KR
GCF_039640425.1	Complete Genome	no			0	80
GCF_039640435.1	Complete Genome	no		NC_003143:2135000:105000	1	5M
GCF_039640445.1	Complete Genome	no			0	2
GCF_039640455.1	Complete Genome	no		NC_003143:2135000:105000	1	8
GCF_039640465.1	Complete Genome	no		NC_003143:2135000:105000	1	197_15
GCF_039640475.1	Complete Genome	no		NC_003143:2135000:105000	1	177
GCF_039640485.1	Complete Genome	no			0	107
GCF_039703215.1	Complete Genome	no	NC_003132		0	AKS2022HT87_B
GCF_039703785.1	Complete Genome	no			0	AKS2022HT87_A
GCF_041409745.1	Complete Genome	no	NC_003132	NC_003143:1310000:10000	1	6/69_pPCP1-
GCF_041410195.1	Complete Genome	no		NC_003143:1310000:10000	1	6/69
GCF_041937415.1	Chromosome	no	NC_003131	NC_003143:1310000:10000;NC_003143:795000:15000	2	KIM6+
GCF_046895365.1	Complete Genome	no			0	7
GCF_046895375.1	Complete Genome	no		NC_003143:1310000:10000;NC_003143:2135000:105000;NC_003143:795000:15000	3	2506
GCF_046895395.1	Complete Genome	no		NC_003143:795000:15000	1	2
GCF_046895405.1	Complete Genome	no	NC_003132		0	CMCCK110001
GCF_046895415.1	Complete Genome	no	NC_003132	NC_003143:2135000:105000;NC_003143:685000:15000;NC_003143:795000:15000	3	CMCC92010
GCF_046895425.1	Complete Genome	no	NC_003131	NC_003143:2760000:10000	1	CMCC05009

accession	assembly level	laboratory named	replicons absent	big blocks 5kb	n big blocks	strain
GCF_046895435.1	Complete Genome	no			0	CMCC640047
GCF_046895445.1	Complete Genome	no			0	CMCC114001
GCF_046895455.1	Complete Genome	no		NC_003143:1895000:30000;NC_003143:3050000:15000;NC_003143:3275000:10000	3	9
GCF_046895465.1	Complete Genome	no			0	19098
GCF_046895475.1	Complete Genome	no		NC_003143:2135000:105000	1	2330
GCF_046895485.1	Complete Genome	no			0	19042
GCF_046895495.1	Complete Genome	no			0	71038
GCF_046895505.1	Complete Genome	no			0	71037
GCF_046895515.1	Complete Genome	no			0	71031
GCF_046895525.1	Complete Genome	no			0	71034
GCF_046895535.1	Chromosome	no			0	71025
GCF_046895545.1	Complete Genome	no			0	19022
GCF_046895555.1	Complete Genome	no		NC_003143:2135000:105000	1	770432
GCF_046895565.1	Complete Genome	no			0	770429
GCF_046895575.1	Complete Genome	no			0	720330
GCF_046895585.1	Complete Genome	no		NC_003143:2365000:25000;NC_003143:795000:15000	2	F020032
GCF_046895595.1	Complete Genome	no			0	71008
GCF_046895605.1	Complete Genome	no		NC_003143:2365000:25000;NC_003143:795000:15000	2	129024
GCF_046895615.1	Complete Genome	no		NC_003143:2135000:105000;NC_003143:795000:15000	2	140068
GCF_046895625.1	Complete Genome	no		NC_003143:795000:15000	1	140037
GCF_046895635.1	Complete Genome	no		NC_003143:2135000:105000;NC_003143:795000:15000	2	140020
GCF_046895645.1	Complete Genome	no		NC_003143:2365000:25000;NC_003143:795000:15000	2	140011
GCF_046895655.1	Complete Genome	no		NC_003143:795000:15000	1	120070
GCF_046895665.1	Complete Genome	no		NC_003143:2135000:105000;NC_003143:795000:15000	2	140003
GCF_046895675.1	Complete Genome	no		NC_003143:795000:15000	1	160009
GCF_046895685.1	Complete Genome	no		NC_003143:795000:15000	1	147014
GCF_046895695.1	Complete Genome	no		NC_003143:2135000:105000;NC_003143:795000:15000	2	146004
GCF_046895705.1	Complete Genome	no		NC_003143:2135000:105000;NC_003143:795000:15000	2	90028
GCF_046895715.1	Complete Genome	no		NC_003143:2135000:105000;NC_003143:795000:15000	2	99008
GCF_046895725.1	Complete Genome	no		NC_003143:2135000:105000;NC_003143:795000:15000	2	99063
GCF_046895735.1	Complete Genome	no		NC_003143:2135000:105000;NC_003143:795000:15000	2	90054

accession	assembly level	laboratory named	replicons absent	big blocks 5kb	n big blocks	strain
GCF_046895745.1	Complete Genome	no		NC_003143:2135000:105000; NC_003143:795000:15000	2	98008
GCF_046895755.1	Complete Genome	no		NC_003143:2135000:105000; NC_003143:795000:15000	2	99002
GCF_046895765.1	Complete Genome	no		NC_003143:795000:15000	1	74002
GCF_046895775.1	Complete Genome	no		NC_003143:2135000:105000; NC_003143:795000:15000	2	124027
GCF_046895785.1	Complete Genome	no		NC_003143:795000:15000	1	47002
GCF_046895795.1	Complete Genome	no		NC_003143:2135000:105000; NC_003143:795000:15000	2	40027
GCF_046895805.1	Complete Genome	no		NC_003143:2135000:105000; NC_003143:795000:15000	2	47005
GCF_046895815.1	Complete Genome	no		NC_003143:795000:15000	1	96011
GCF_046895825.1	Complete Genome	no		NC_003143:795000:15000	1	83007
GCF_046895835.1	Complete Genome	no		NC_003143:2135000:105000; NC_003143:795000:15000	2	90057
GCF_046895845.1	Complete Genome	no		NC_003143:795000:15000	1	95037
GCF_046895855.1	Complete Genome	no		NC_003143:2135000:105000; NC_003143:795000:15000	2	95033
GCF_046895865.1	Complete Genome	no		NC_003143:795000:15000	1	70016
GCF_046895875.1	Complete Genome	no		NC_003143:795000:15000	1	55019
GCF_046895885.1	Complete Genome	no		NC_003143:795000:15000	1	55014
GCF_046895895.1	Complete Genome	no		NC_003143:795000:15000	1	50008
GCF_046895905.1	Complete Genome	no		NC_003143:795000:15000	1	50018
GCF_046895915.1	Complete Genome	no		NC_003143:2135000:105000; NC_003143:795000:15000	2	96005
GCF_046895925.1	Complete Genome	no		NC_003143:795000:15000	1	60007
GCF_046895935.1	Complete Genome	no		NC_003143:795000:15000	1	54009
GCF_046895945.1	Complete Genome	no		NC_003143:795000:15000	1	94012
GCF_046895955.1	Complete Genome	no		NC_003143:795000:15000	1	50010
GCF_046895965.1	Complete Genome	no		NC_003143:795000:15000	1	51012
GCF_046895975.1	Complete Genome	no		NC_003143:795000:15000	1	94007
GCF_046895985.1	Complete Genome	no		NC_003143:795000:15000	1	51022
GCF_052373935.1	Complete Genome	no		NC_003143:1110000:30000	1	38_YP25_SZ
GCF_052373945.1	Complete Genome	no		NC_003143:1110000:30000	1	34_YP28_SZ
GCF_052373955.1	Complete Genome	no		NC_003143:1110000:30000	1	8_YP55_NP
GCF_052373995.1	Complete Genome	no		NC_003143:1110000:30000	1	23_YP50_MANG
GCF_052374015.1	Complete Genome	no		NC_003143:1310000:10000;NC_003143:2745000:40000;NC_003143:615000:80000	3	39_YP13_TLH
GCF_052374055.1	Complete Genome	no		NC_003143:1310000:10000;NC_003143:2745000:40000;NC_003143:615000:80000	3	31_YP22_SZ

accession	assembly level	laboratory named	replicons absent	big blocks 5kb	n big blocks	strain
GCF_052374075.1	Complete Genome	no		NC_003143:1310000:10000;NC_003143:2745000:40000;NC_003143:615000:80000	3	35_YP14_TLH
GCF_052374095.1	Complete Genome	no		NC_003143:1310000:10000;NC_003143:2745000:40000;NC_003143:615000:80000	3	28_YP29_SZ
GCF_052374115.1	Complete Genome	no		NC_003143:1310000:10000;NC_003143:2745000:40000;NC_003143:615000:80000	3	27_YP31_SZ
GCF_052374135.1	Complete Genome	no		NC_003143:1310000:10000;NC_003143:2745000:40000;NC_003143:615000:80000	3	26_YP30_SZ
GCF_052374155.1	Complete Genome	no		NC_003143:1310000:10000;NC_003143:2745000:40000;NC_003143:615000:80000	3	43_YP6_UE
GCF_052374175.1	Complete Genome	no		NC_003143:1310000:10000;NC_003143:2745000:40000;NC_003143:615000:80000	3	42_YP10_UE
GCF_052374195.1	Complete Genome	no		NC_003143:1310000:10000;NC_003143:2745000:40000;NC_003143:615000:80000	3	33_YP16_UE
GCF_052374215.1	Complete Genome	no		NC_003143:1310000:10000;NC_003143:2745000:40000;NC_003143:615000:80000	3	58_YP9_TK
GCF_052374235.1	Complete Genome	no		NC_003143:1310000:10000;NC_003143:2745000:40000;NC_003143:615000:80000	3	56_YP12_TK
GCF_052374255.1	Complete Genome	no		NC_003143:1310000:10000;NC_003143:2745000:40000;NC_003143:615000:80000	3	79_YP9_MOK
GCF_052374275.1	Complete Genome	no		NC_003143:1310000:10000;NC_003143:2745000:40000;NC_003143:615000:80000	3	76_YP20_MOK
GCF_052374295.1	Complete Genome	no		NC_003143:1310000:10000;NC_003143:2745000:40000;NC_003143:615000:80000	3	55_YP20_MOK
GCF_052374315.1	Complete Genome	no		NC_003143:1310000:10000;NC_003143:2745000:40000;NC_003143:615000:80000	3	32_YP32_MOK
GCF_052374335.1	Complete Genome	no		NC_003143:1310000:10000;NC_003143:2745000:40000;NC_003143:615000:80000	3	52_YP18_PAK
GCF_052374355.1	Complete Genome	no		NC_003143:1310000:10000;NC_003143:2745000:40000;NC_003143:615000:80000	3	51_YP1_PAK

accession	assembly level	laboratory named	replicons absent	big blocks 5kb	n big blocks	strain
GCF_052374375.1	Complete Genome	no		NC_003143:1310000:10000;NC_003143:2745000:40000;NC_003143:6150000:80000	3	49_YP5_PAK
GCF_052374395.1	Complete Genome	no		NC_003143:1310000:10000;NC_003143:2745000:40000;NC_003143:6150000:80000	3	47_YP36_PAK
GCF_052374415.1	Complete Genome	no		NC_003143:1310000:10000;NC_003143:2745000:40000;NC_003143:6150000:80000	3	46_YP29_PAK
GCF_052374445.1	Complete Genome	no		NC_003143:1310000:10000;NC_003143:2745000:40000;NC_003143:6150000:80000	3	41_YP26_PAK
GCF_052374465.1	Complete Genome	no		NC_003143:1310000:10000;NC_003143:2745000:40000;NC_003143:6150000:80000	3	75_YP26_PB
GCF_052374495.1	Complete Genome	no		NC_003143:1310000:10000;NC_003143:2745000:40000;NC_003143:6150000:80000	3	68_YP15_PB
GCF_052374515.1	Complete Genome	no		NC_003143:1310000:10000;NC_003143:2745000:40000;NC_003143:6150000:80000	3	67_YP34_PB
GCF_052374535.1	Complete Genome	no		NC_003143:1310000:10000;NC_003143:2745000:40000;NC_003143:6150000:80000	3	62_YP33_PB
GCF_052374555.1	Complete Genome	no		NC_003143:1310000:10000;NC_003143:2745000:40000;NC_003143:6150000:80000	3	61_YP3_PB
GCF_052374575.1	Complete Genome	no		NC_003143:1310000:10000;NC_003143:2745000:40000;NC_003143:6150000:80000	3	50_YP23_PB
GCF_052374595.1	Complete Genome	no		NC_003143:1310000:10000;NC_003143:2745000:40000;NC_003143:6150000:80000	3	40_YP4_PB
GCF_052374615.1	Complete Genome	no		NC_003143:1310000:10000;NC_003143:2745000:40000;NC_003143:6150000:80000	3	63_YP2_PB
GCF_052374635.1	Complete Genome	no		NC_003143:1310000:10000;NC_003143:2745000:40000;NC_003143:6150000:80000	3	30_YP17_PB
GCF_052374655.1	Complete Genome	no		NC_003143:1310000:10000;NC_003143:2745000:40000;NC_003143:6150000:80000	3	60_YP24_IM
GCF_052374675.1	Complete Genome	no		NC_003143:1310000:10000;NC_003143:2745000:40000;NC_003143:6150000:80000	3	59_YP21_IM

accession	assembly level	laboratory named	replicons absent	big blocks 5kb	n big blocks	strain
GCF_052374695.1	Complete Genome	no		NC_003143:1310000:10000;NC_003143:2745000:40000;NC_003143:615000:80000	3	54_YP2_IM
GCF_052374715.1	Complete Genome	no		NC_003143:1310000:10000;NC_003143:2745000:40000;NC_003143:615000:80000	3	25_YP73_IM
GCF_052374735.1	Complete Genome	no		NC_003143:1310000:10000;NC_003143:2745000:40000;NC_003143:615000:80000	3	24_YP70_IM
GCF_052374755.1	Complete Genome	no		NC_003143:1310000:10000;NC_003143:2745000:40000;NC_003143:615000:80000	3	23_YP72_IM
GCF_052374775.1	Complete Genome	no		NC_003143:1310000:10000;NC_003143:2745000:40000;NC_003143:615000:80000	3	22_YP66_IM
GCF_052374795.1	Complete Genome	no		NC_003143:1310000:10000;NC_003143:2745000:40000;NC_003143:615000:80000	3	21_YP63_IM
GCF_052374815.1	Complete Genome	no		NC_003143:1310000:10000;NC_003143:2745000:40000;NC_003143:615000:80000	3	20_YP71_IM
GCF_052374855.1	Complete Genome	no		NC_003143:1310000:10000;NC_003143:2745000:40000;NC_003143:615000:80000	3	18_YP64_IM
GCF_052374875.1	Complete Genome	no		NC_003143:1310000:10000;NC_003143:2745000:40000;NC_003143:615000:80000	3	17_YP69_IM
GCF_052374895.1	Complete Genome	no		NC_003143:1310000:10000;NC_003143:2745000:40000;NC_003143:615000:80000	3	16_YP53_IM
GCF_052374915.1	Complete Genome	no		NC_003143:1310000:10000;NC_003143:2745000:40000;NC_003143:615000:80000	3	15_YP61_IM
GCF_052374935.1	Complete Genome	no		NC_003143:1310000:10000;NC_003143:2745000:40000;NC_003143:615000:80000	3	14_YP75_IM
GCF_052374955.1	Complete Genome	no		NC_003143:1310000:10000;NC_003143:2745000:40000;NC_003143:615000:80000	3	13_YP56_IM
GCF_052374975.1	Complete Genome	no		NC_003143:1310000:10000;NC_003143:2745000:40000;NC_003143:615000:80000	3	12_YP54_IM
GCF_052374995.1	Complete Genome	no		NC_003143:1310000:10000;NC_003143:2745000:40000;NC_003143:615000:80000	3	11_YP51_IM

accession	assembly level	laboratory named	replicons absent	big blocks 5kb	n big blocks	strain
GCF_052375015.1	Complete Genome	no		NC_003143:1310000:10000;NC_003143:2745000:40000;NC_003143:615000:80000	3	10_YP59_IM
GCF_052375035.1	Complete Genome	no		NC_003143:1310000:10000;NC_003143:2745000:40000;NC_003143:615000:80000	3	9_YP68_IM
GCF_052375045.1	Complete Genome	no		NC_003143:1310000:10000;NC_003143:2745000:40000;NC_003143:615000:80000	3	7_YP60_IM
GCF_052375075.1	Complete Genome	no		NC_003143:1310000:10000;NC_003143:2745000:40000;NC_003143:615000:80000	3	82_YP1_IM
GCF_052375095.1	Complete Genome	no		NC_003143:1310000:10000;NC_003143:2745000:40000;NC_003143:615000:80000	3	81_YP39_IM
GCF_052375105.1	Complete Genome	no		NC_003143:1310000:10000;NC_003143:2745000:40000;NC_003143:615000:80000	3	80_YP4_IM
GCF_052375135.1	Complete Genome	no		NC_003143:1310000:10000;NC_003143:2745000:40000;NC_003143:615000:80000	3	78_YP23_IM
GCF_052375155.1	Complete Genome	no		NC_003143:1310000:10000;NC_003143:2745000:40000;NC_003143:615000:80000	3	77_YP25_IM
GCF_052375175.1	Complete Genome	no		NC_003143:1310000:10000;NC_003143:2745000:40000;NC_003143:615000:80000	3	71_YP7_IM
GCF_052375195.1	Complete Genome	no		NC_003143:1310000:10000;NC_003143:2745000:40000;NC_003143:615000:80000	3	70_YP27_IM
GCF_052375235.1	Complete Genome	no		NC_003143:1310000:10000;NC_003143:2745000:40000;NC_003143:615000:80000	3	44_YP7_NP
GCF_052375255.1	Complete Genome	no		NC_003143:1310000:10000;NC_003143:2745000:40000;NC_003143:615000:80000	3	98_YP8_KK
GCF_052375265.1	Complete Genome	no		NC_003143:1310000:10000;NC_003143:2745000:40000;NC_003143:615000:80000	3	97_YP1_KK
GCF_052375295.1	Complete Genome	no		NC_003143:1310000:10000;NC_003143:2745000:40000;NC_003143:615000:80000	3	96_YP2_KK
GCF_052375315.1	Complete Genome	no		NC_003143:1310000:10000;NC_003143:2745000:40000;NC_003143:615000:80000	3	94_YP6_KK

accession	assembly level	laboratory named	replicons absent	big blocks 5kb	n big blocks	strain
GCF_052375335.1	Complete Genome	no		NC_003143:1310000:10000;NC_003143:2745000:40000;NC_003143:615000:80000	3	95_YP11_KK
GCF_052375345.1	Complete Genome	no		NC_003143:1310000:10000;NC_003143:2745000:40000;NC_003143:615000:80000	3	93_YP24_KK
GCF_052375375.1	Complete Genome	no		NC_003143:1310000:10000;NC_003143:2745000:40000;NC_003143:615000:80000	3	92_YP31_KK
GCF_052375395.1	Complete Genome	no		NC_003143:1310000:10000;NC_003143:2745000:40000;NC_003143:615000:80000	3	74_YP19_KK
GCF_052375415.1	Complete Genome	no		NC_003143:1310000:10000;NC_003143:2745000:40000;NC_003143:615000:80000	3	73_YP21_KK
GCF_052375435.1	Complete Genome	no		NC_003143:1310000:10000;NC_003143:2745000:40000;NC_003143:615000:80000	3	64_YP37_KK
GCF_052375455.1	Complete Genome	no		NC_003143:1310000:10000;NC_003143:2745000:40000;NC_003143:615000:80000	3	65_YP4_KK
GCF_052375475.1	Complete Genome	no		NC_003143:1310000:10000;NC_003143:2745000:40000;NC_003143:615000:80000	3	6_YP52_KK
GCF_052375495.1	Complete Genome	no		NC_003143:1310000:10000;NC_003143:2745000:40000;NC_003143:615000:80000	3	29_YP8_KK
GCF_052375515.1	Complete Genome	no		NC_003143:1310000:10000;NC_003143:2745000:40000;NC_003143:615000:80000	3	72_YP5_PKK
GCF_052375535.1	Complete Genome	no		NC_003143:1310000:10000;NC_003143:2745000:40000;NC_003143:615000:80000	3	87_YP18_PKK
GCF_052375555.1	Complete Genome	no		NC_003143:1310000:10000;NC_003143:2745000:40000;NC_003143:615000:80000	3	5_YP57_PKK
GCF_052375615.1	Complete Genome	no		NC_003143:1310000:10000;NC_003143:2745000:40000;NC_003143:615000:80000	3	4_YP65_PKK
GCF_052375635.1	Complete Genome	no		NC_003143:1310000:10000;NC_003143:2745000:40000;NC_003143:615000:80000	3	91_YP6_ADT
GCF_052375655.1	Complete Genome	no		NC_003143:1310000:10000;NC_003143:2745000:40000;NC_003143:615000:80000	3	89_YP5_ADT

accession	assembly level	laboratory named	replicons absent	big blocks 5kb	n big blocks	strain
GCF_052375675.1	Complete Genome	no		NC_003143:1310000:10000;NC_003143:2745000:40000;NC_003143:6150000:80000	3	88_YP10_ADT
GCF_052375695.1	Complete Genome	no		NC_003143:1310000:10000;NC_003143:2745000:40000;NC_003143:6150000:80000	3	90_YP32_ADT
GCF_052375715.1	Complete Genome	no		NC_003143:1310000:10000;NC_003143:2745000:40000;NC_003143:6150000:80000	3	85_YP3_ADT
GCF_052375735.1	Complete Genome	no		NC_003143:1310000:10000;NC_003143:2745000:40000;NC_003143:6150000:80000	3	86_YP12_ADT
GCF_052375755.1	Complete Genome	no		NC_003143:1310000:10000;NC_003143:2745000:40000;NC_003143:6150000:80000	3	84_YP17_ADT
GCF_052375775.1	Complete Genome	no		NC_003143:1310000:10000;NC_003143:2745000:40000;NC_003143:6150000:80000	3	83_YP13_ADT
GCF_052375795.1	Complete Genome	no		NC_003143:1310000:10000;NC_003143:2745000:40000;NC_003143:6150000:80000	3	3_YP58_ADT
GCF_052375815.1	Complete Genome	no		NC_003143:1310000:10000;NC_003143:2745000:40000;NC_003143:6150000:80000	3	69_YP30_ADT
GCF_052375835.1	Complete Genome	no		NC_003143:1310000:10000;NC_003143:2745000:40000;NC_003143:6150000:80000	3	2_YP67_ADT
GCF_052375915.1	Complete Genome	no		NC_003143:1310000:10000;NC_003143:2745000:40000;NC_003143:6150000:80000	3	1_YP62_ADT
GCF_052446015.1	Complete Genome	no		NC_003143:1310000:10000;NC_003143:2745000:40000;NC_003143:6150000:80000	3	45_YP8_PAK
GCF_053834845.1	Complete Genome	no		NC_003143:2370000:35000	1	115
GCF_053834855.1	Complete Genome	no			0	166
GCF_053834865.1	Complete Genome	no		NC_003143:1310000:10000;NC_003143:7950000:15000	2	2115
GCF_053834875.1	Complete Genome	no		NC_003143:2370000:35000	1	109
GCF_061402485.1	Complete Genome	no	NC_003132		0	YN24HQzi2

source: results/qc/deletion_profile_all.tsv + assembly_metadata.tsv + lab_strain_inventory.tsv

Table S6. Pseudogene-deletion entanglement in the curated CO92 annotation against the automated annotation.

replicon	ncbi start	ncbi end	ncbi span	ncbi gene	ncbi product	bakta span	bakta frac of ncbi	bakta gene	category
NC_003143	55510	56126	617	YPO_RS01245		624	1.011	yadS	full_length_bakta_CDS

replicon	ncbi start	ncbi end	ncbi span	ncbi gene	ncbi product	bakta span	bakta frac of ncbi	bakta gene	category
NC_003143	72571	73505	935	YPO_RS01345		501	0.536	KIGGAN_00360	shorter_bakta_CDS
NC_003143	85128	85460	333	YPO_RS01410		0	0.0		no_bakta_CDS
NC_003143	99525	100277	753	glpX		660	0.876	glpX	full_length_bakta_CDS
NC_003143	100198	101355	1158	glpK		1170	1.01	glpK	full_length_bakta_CDS
NC_003143	104291	105580	1290	YPO_RS01510		1218	0.944	KIGGAN_00535	full_length_bakta_CDS
NC_003143	107532	108365	834	YPO_RS01525		906	1.086	KIGGAN_00550	full_length_bakta_CDS
NC_003143	122247	123406	1160	metB		990	0.853	metB	full_length_bakta_CDS
NC_003143	134066	136470	2405	malP		2049	0.852	malP	full_length_bakta_CDS
NC_003143	189518	190051	534	argD		606	1.135	KIGGAN_00900	full_length_bakta_CDS
NC_003143	191999	192694	696	argD		741	1.065	KIGGAN_00915	full_length_bakta_CDS
NC_003143	206063	207981	1919	YPO_RS01950		1386	0.722	KIGGAN_00995	shorter_bakta_CDS
NC_003143	273676	273747	72	YPO_RS21960		0	0.0		no_bakta_CDS
NC_003143	307028	307111	84	YPO_RS22480		0	0.0		no_bakta_CDS
NC_003143	344496	344852	357	YPO_RS02685		0	0.0		no_bakta_CDS
NC_003143	406860	407310	451	YPO_RS02950		264	0.585	tnp	shorter_bakta_CDS
NC_003143	431515	432502	988	lsrR		627	0.635	sorC	shorter_bakta_CDS
NC_003143	458193	459511	1319	deoA		780	0.591	KIGGAN_02280	shorter_bakta_CDS
NC_003143	468068	469731	1664	ettA		1077	0.647	ettA	shorter_bakta_CDS
NC_003143	544987	547137	2151	YPO_RS22635		1590	0.739	KIGGAN_02665	shorter_bakta_CDS
NC_003143	556834	556902	69	YPO_RS21980		0	0.0		no_bakta_CDS
NC_003143	570251	571183	933	YPO_RS03630		957	1.026	araJ	full_length_bakta_CDS
NC_003143	573140	573391	252	YPO_RS03645		270	1.071	KIGGAN_02770	full_length_bakta_CDS
NC_003143	612115	612207	93	YPO_RS21770		0	0.0		no_bakta_CDS
NC_003143	614923	615027	105	YPO_RS03830		180	1.714	KIGGAN_02985	full_length_bakta_CDS
NC_003143	643778	644817	1040	YPO_RS03955		984	0.946	terY	full_length_bakta_CDS
NC_003143	687471	688066	596	YPO_RS22645		627	1.052	yheO	full_length_bakta_CDS
NC_003143	691802	691948	147	YPO_RS22650		0	0.0		no_bakta_CDS
NC_003143	735996	738340	2345	YPO_RS04345		1548	0.66	ygiQ	shorter_bakta_CDS
NC_003143	766793	768890	2098	YPO_RS04495		1302	0.621	flhA	shorter_bakta_CDS
NC_003143	785180	785296	117	YPO_RS22030		189	1.615	KIGGAN_03795	full_length_bakta_CDS
NC_003143	786005	786622	618	YPO_RS04610		594	0.961	flgF	full_length_bakta_CDS

replicon	ncbi start	ncbi end	ncbi span	ncbi gene	ncbi product	bakta span	bakta frac of ncbi	bakta gene	category
NC_003143	808221	808689	469	YPO_RS04730		0	0.0	KIGGAN_03930	no_bakta_CDS
NC_003143	858886	859281	396	YPO_RS22390		402	1.015	KIGGAN_04075	full_length_bakta_CDS
NC_003143	867807	867899	93	YPO_RS22045		165	1.774	yaiL	full_length_bakta_CDS
NC_003143	890464	890574	111	YPO_RS22485		543	4.892	pulG	full_length_bakta_CDS
NC_003143	908636	908698	63	YPO_RS22050		0	0.0		no_bakta_CDS
NC_003143	951571	951756	186	YPO_RS22055		0	0.0	yycO	no_bakta_CDS
NC_003143	990362	990457	96	YPO_RS05450		0	0.0		no_bakta_CDS
NC_003143	1009670	1009738	69	YPO_RS21790		0	0.0		no_bakta_CDS
NC_003143	1047310	1048043	734	YPO_RS05680		531	0.723	yggN	shorter_bakta_CDS
NC_003143	1064254	1064559	306	YPO_RS05750		309	1.01	KIGGAN_05040	full_length_bakta_CDS
NC_003143	1070947	1071520	574	YPO_RS05780		0	0.0	tnp	no_bakta_CDS
NC_003143	1087623	1087718	96	YPO_RS22065		156	1.625	KIGGAN_05125	full_length_bakta_CDS
NC_003143	1088261	1088619	359	YPO_RS05840		264	0.735	insA	shorter_bakta_CDS
NC_003143	1097538	1099309	1772	YPO_RS05885		1071	0.604	iucC	shorter_bakta_CDS
NC_003143	1133325	1133429	105	YPO_RS22675		87	0.829	KIGGAN_05270	full_length_bakta_CDS
NC_003143	1147884	1148060	177	YPO_RS22505		0	0.0	pulJ	no_bakta_CDS
NC_003143	1232065	1232283	219	YPO_RS06405		234	1.068	KIGGAN_05740	full_length_bakta_CDS
NC_003143	1234224	1234340	117	YPO_RS06420		177	1.513	KIGGAN_05755	full_length_bakta_CDS
NC_003143	1234755	1235071	317	YPO_RS06425		0	0.0		no_bakta_CDS
NC_003143	1238047	1238364	318	YPO_RS22685		543	1.708	KIGGAN_05780	full_length_bakta_CDS
NC_003143	1280019	1280114	96	YPO_RS06670		0	0.0		no_bakta_CDS
NC_003143	1282023	1283092	1070	galM		897	0.838	galM	full_length_bakta_CDS
NC_003143	1320482	1321174	693	YPO_RS06865		801	1.156	KIGGAN_06230	full_length_bakta_CDS
NC_003143	1345498	1346430	933	osmF		468	0.502	osmF	shorter_bakta_CDS
NC_003143	1377189	1379881	2693	rcsD		1953	0.725	rcsD	shorter_bakta_CDS
NC_003143	1395966	1396615	650	YPO_RS07180		612	0.942	araD	full_length_bakta_CDS
NC_003143	1403916	1404002	87	YPO_RS22690		0	0.0		no_bakta_CDS
NC_003143	1413775	1413864	90	YPO_RS21800		93	1.033	KIGGAN_06680	full_length_bakta_CDS
NC_003143	1502466	1502753	288	YPO_RS07660		303	1.052	KIGGAN_07085	full_length_bakta_CDS
NC_003143	1504695	1505156	462	YPO_RS07675		519	1.123	KIGGAN_07100	full_length_bakta_CDS
NC_003143	1585135	1585264	130	YPO_RS22700		0	0.0		no_bakta_CDS
NC_003143	1619438	1620412	975	YPO_RS08120		1029	1.055	fabB	full_length_bakta_CDS

replicon	ncbi start	ncbi end	ncbi span	ncbi gene	ncbi product	bakta span	bakta frac of ncbi	bakta gene	category
NC_003143	1622369	1622671	303	YPO_RS08135		375	1.238	fabB	full_length_bakta_CDS
NC_003143	1639931	1640424	494	YPO_RS08235		288	0.583	KIGGAN_07675	shorter_bakta_CDS
NC_003143	1640827	1642937	2111	cnfy		1170	0.554	KIGGAN_07690	shorter_bakta_CDS
NC_003143	1674953	1675823	871	YPO_RS08375		0	0.0		no_bakta_CDS
NC_003143	1773208	1773404	197	YPO_RS08775		0	0.0		no_bakta_CDS
NC_003143	1811235	1811378	144	YPO_RS08935		0	0.0		no_bakta_CDS
NC_003143	1827608	1827676	69	YPO_RS22110		0	0.0		no_bakta_CDS
NC_003143	1853654	1854569	916	nagK		771	0.842	nagK	full_length_bakta_CDS
NC_003143	1884574	1886308	1735	YPO_RS09275		1320	0.761	KIGGAN_08765	shorter_bakta_CDS
NC_003143	1892420	1892771	352	flhD		258	0.733	flhD	shorter_bakta_CDS
NC_003143	1912878	1914488	1611	YPO_RS09360		1218	0.756	KIGGAN_08850	shorter_bakta_CDS
NC_003143	1919412	1922043	2632	YPO_RS09390		2211	0.84	hia	full_length_bakta_CDS
NC_003143	2004462	2005917	1456	YPO_RS09780		1377	0.946	tlyC	full_length_bakta_CDS
NC_003143	2038736	2039584	849	inv		1011	1.191	KIGGAN_09470	full_length_bakta_CDS
NC_003143	2040296	2042350	2055	inv		2073	1.009	KIGGAN_09485	full_length_bakta_CDS
NC_003143	2047414	2048028	615	flgF		630	1.024	flgF	full_length_bakta_CDS
NC_003143	2048012	2048722	711	flgG		699	0.983	flgG	full_length_bakta_CDS
NC_003143	2077749	2077919	171	YPO_RS10145		0	0.0		no_bakta_CDS
NC_003143	2093287	2093355	69	YPO_RS22125		0	0.0		no_bakta_CDS
NC_003143	2120692	2121209	518	YPO_RS10375		306	0.591	KIGGAN_09930	shorter_bakta_CDS
NC_003143	2124367	2125565	1199	YPO_RS10410		861	0.718	KIGGAN_09965	shorter_bakta_CDS
NC_003143	2127296	2127622	327	YPO_RS10425		0	0.0		no_bakta_CDS
NC_003143	2127676	2127875	200	YPO_RS22130		105	0.525	KIGGAN_09990	shorter_bakta_CDS
NC_003143	2139040	2139967	928	YPO_RS22135		555	0.598	KIGGAN_10075	shorter_bakta_CDS
NC_003143	2235390	2236265	876	YPO_RS10830		948	1.082	KIGGAN_10415	full_length_bakta_CDS
NC_003143	2238217	2238429	213	YPO_RS10845		0	0.0	KIGGAN_10430	no_bakta_CDS
NC_003143	2281003	2281928	926	YPO_RS11040		510	0.551	tra5	shorter_bakta_CDS
NC_003143	2294432	2294860	429	YPO_RS11110		423	0.986	dauA	full_length_bakta_CDS
NC_003143	2296817	2298088	1272	dauA		1278	1.005	dauA	full_length_bakta_CDS
NC_003143	2303129	2303890	762	YPO_RS11155		885	1.161	KIGGAN_10765	full_length_bakta_CDS
NC_003143	2304595	2304906	312	YPO_RS11165		327	1.048	KIGGAN_10780	full_length_bakta_CDS

replicon	ncbi start	ncbi end	ncbi span	ncbi gene	ncbi product	bakta span	bakta frac of ncbi	bakta gene	category
NC_003143	2312549	2313187	639	YPO_RS22145		1611	2.521	yceH	full_length_bakta_CDS
NC_003143	2375102	2375656	555	YPO_RS11510		579	1.043	KIGGAN_11150	full_length_bakta_CDS
NC_003143	2376974	2377915	942	YPO_RS11520		975	1.035	KIGGAN_11160	full_length_bakta_CDS
NC_003143	2416649	2417838	1190	YPO_RS11730		1203	1.011	araJ	full_length_bakta_CDS
NC_003143	2481001	2482592	1592	trpD		999	0.628	trpD	shorter_bakta_CDS
NC_003143	2547842	2549140	1299	YPO_RS12325		1119	0.861	araJ	full_length_bakta_CDS
NC_003143	2567674	2568174	501	YPO_RS22175		0	0.0		no_bakta_CDS
NC_003143	2568132	2569595	1464	YPO_RS12445		0	0.0		no_bakta_CDS
NC_003143	2581241	2583636	2396	YPO_RS22720		1386	0.578	KIGGAN_12125	shorter_bakta_CDS
NC_003143	2713182	2714645	1464	YPO_RS13070		0	0.0	selO	no_bakta_CDS
NC_003143	2786419	2788503	2085	YPO_RS13430		2034	0.976	KIGGAN_13115	full_length_bakta_CDS
NC_003143	2804577	2804645	69	YPO_RS21860		0	0.0		no_bakta_CDS
NC_003143	2866561	2867060	500	nuoE		564	1.128	nuoE	full_length_bakta_CDS
NC_003143	2883273	2883341	69	YPO_RS22205		0	0.0		no_bakta_CDS
NC_003143	2900557	2901542	986	iolG		510	0.517	KIGGAN_13580	shorter_bakta_CDS
NC_003143	2960480	2963144	2665	YPO_RS14140		2103	0.789	KIGGAN_13880	shorter_bakta_CDS
NC_003143	2968124	2968384	261	YPO_RS14175		264	1.011	KIGGAN_13920	full_length_bakta_CDS
NC_003143	2968454	2968864	411	YPO_RS14180		483	1.175	KIGGAN_13925	full_length_bakta_CDS
NC_003143	2971684	2971752	69	YPO_RS22735		0	0.0		no_bakta_CDS
NC_003143	2983593	2983808	216	YPO_RS14255		231	1.069	KIGGAN_14010	full_length_bakta_CDS
NC_003143	2996498	2997464	967	YPO_RS14320		834	0.862	ureD	full_length_bakta_CDS
NC_003143	3026599	3027255	657	pxpB		507	0.772	KIGGAN_14235	shorter_bakta_CDS
NC_003143	3050889	3051674	786	tssL		837	1.065	tssL	full_length_bakta_CDS
NC_003143	3091143	3091220	78	YPO_RS22225		0	0.0		no_bakta_CDS
NC_003143	3114458	3115046	589	yfcE		561	0.952	yfcE	full_length_bakta_CDS
NC_003143	3118571	3120103	1533	pbpC		1392	0.908	pbpC	full_length_bakta_CDS
NC_003143	3121234	3122090	857	YPO_RS14925		582	0.679	KIGGAN_14715	shorter_bakta_CDS
NC_003143	3128111	3128461	351	YPO_RS14970		309	0.88	KIGGAN_14760	full_length_bakta_CDS
NC_003143	3149504	3149773	270	YPO_RS15065		0	0.0		no_bakta_CDS
NC_003143	3157528	3158717	1190	fliB		834	0.701	fliB	shorter_bakta_CDS
NC_003143	3188386	3189767	1382	baeS		1116	0.808	baeS	full_length_bakta_CDS

replicon	ncbi start	ncbi end	ncbi span	ncbi gene	ncbi product	bakta span	bakta frac of ncbi	bakta gene	category
NC_003143	3252520	3253420	901	yhfZ		597	0.663	gntR	shorter_bakta_CDS
NC_003143	3277040	3277636	597	YPO_RS15630		633	1.06	dotU	full_length_bakta_CDS
NC_003143	3356083	3357302	1220	YPO_RS16010		1032	0.846	KIGGAN_15850	full_length_bakta_CDS
NC_003143	3412943	3413059	117	YPO_RS22255		0	0.0	tmcA	no_bakta_CDS
NC_003143	3447128	3448783	1656	ushA		0	0.0	KIGGAN_16300	no_bakta_CDS
NC_003143	3458733	3459697	965	fcl		714	0.74	KIGGAN_16345	shorter_bakta_CDS
NC_003143	3459703	3460825	1123	gmd		639	0.569	KIGGAN_16355	shorter_bakta_CDS
NC_003143	3462103	3463324	1222	wzy		807	0.66	KIGGAN_16370	shorter_bakta_CDS
NC_003143	3470493	3471568	1076	rfbG		996	0.926	rfbG	full_length_bakta_CDS
NC_003143	3483344	3484364	1021	istA		1800	1.763	KIGGAN_16465	full_length_bakta_CDS
NC_003143	3484449	3484724	276	YPO_RS16570		279	1.011	KIGGAN_16470	full_length_bakta_CDS
NC_003143	3557687	3559452	1766	ggt		1515	0.858	ggt	full_length_bakta_CDS
NC_003143	3582249	3582812	564	YPO_RS17005		522	0.926	ubiE	full_length_bakta_CDS
NC_003143	3627211	3628119	909	YPO_RS17180		924	1.017	KIGGAN_17100	full_length_bakta_CDS
NC_003143	3661761	3662693	933	pheA		939	1.006	pheA	full_length_bakta_CDS
NC_003143	3664650	3664877	228	pheA		219	0.961	KIGGAN_17285	full_length_bakta_CDS
NC_003143	3668564	3669908	1345	YPO_RS17380		681	0.506	KIGGAN_17305	shorter_bakta_CDS
NC_003143	3726211	3728307	2097	YPO_RS17655		1584	0.755	KIGGAN_17605	shorter_bakta_CDS
NC_003143	3757428	3758154	727	YPO_RS17815		468	0.644	KIGGAN_17770	shorter_bakta_CDS
NC_003143	3770952	3771068	117	YPO_RS17865		0	0.0		no_bakta_CDS
NC_003143	3811297	3811374	78	YPO_RS22745		0	0.0		no_bakta_CDS
NC_003143	3855341	3856908	1568	YPO_RS18210		1173	0.748	ddpA	shorter_bakta_CDS
NC_003143	3894143	3894220	78	YPO_RS18385		0	0.0		no_bakta_CDS
NC_003143	3952335	3953218	884	YPO_RS18685		942	1.066	eCM4	full_length_bakta_CDS
NC_003143	3993920	3994207	288	hpf		0	0.0	hpf	no_bakta_CDS
NC_003143	4000718	4000924	207	YPO_RS18930		264	1.275	KIGGAN_18930	full_length_bakta_CDS
NC_003143	4000981	4001718	738	YPO_RS18935		744	1.008	KIGGAN_18935	full_length_bakta_CDS
NC_003143	4045510	4045707	198	YPO_RS19125		0	0.0	yafQ	no_bakta_CDS
NC_003143	4045707	4045775	69	YPO_RS22315		0	0.0		no_bakta_CDS
NC_003143	4072548	4072745	198	YPO_RS19250		0	0.0		no_bakta_CDS
NC_003143	4072868	4073038	171	YPO_RS19255		0	0.0		no_bakta_CDS
NC_003143	4122382	4123850	1469	YPO_RS19435		1134	0.772	KIGGAN_19440	shorter_bakta_CDS

replicon	ncbi start	ncbi end	ncbi span	ncbi gene	ncbi product	bakta span	bakta frac of ncbi	bakta gene	category
NC_003143	4136714	4137867	1154	YPO_RS19505		786	0.681	tnp	shorter_bakta_CDS
NC_003143	4137910	4138983	1074	tssF		1227	1.142	tssF	full_length_bakta_CDS
NC_003143	4167025	4167869	845	iclR		498	0.589	iclR	shorter_bakta_CDS
NC_003143	4226995	4227165	171	YPO_RS19900		210	1.228	KIGGAN_19940	full_length_bakta_CDS
NC_003143	4236482	4236790	309	YPO_RS19940		336	1.087	KIGGAN_19980	full_length_bakta_CDS
NC_003143	4238729	4239223	495	YPO_RS19955		540	1.091	tatD	full_length_bakta_CDS
NC_003143	4256544	4257499	956	YPO_RS20035		927	0.97	yfdV	full_length_bakta_CDS
NC_003143	4294294	4294374	81	YPO_RS22325		0	0.0		no_bakta_CDS
NC_003143	4295675	4296866	1192	YPO_RS20225		1053	0.883	KIGGAN_20265	full_length_bakta_CDS
NC_003143	4316509	4316577	69	YPO_RS22760		0	0.0		no_bakta_CDS
NC_003143	4345240	4346731	1492	gppA		1011	0.678	ppx	shorter_bakta_CDS
NC_003143	4367333	4367395	63	YPO_RS22765		483	7.667	KIGGAN_20625	full_length_bakta_CDS
NC_003143	4367874	4367936	63	YPO_RS22770		486	7.714	KIGGAN_20630	full_length_bakta_CDS
NC_003143	4420162	4420323	162	YPO_RS20790		0	0.0	KIGGAN_20835	no_bakta_CDS
NC_003143	4421260	4422681	1422	glpD		1422	1.0	glpD	full_length_bakta_CDS
NC_003143	4427101	4428388	1288	glgC		1431	1.111	glgC	full_length_bakta_CDS
NC_003143	4445467	4445605	139	YPO_RS20835		144	1.036	KIGGAN_20885	full_length_bakta_CDS
NC_003143	4450190	4450420	231	YPO_RS20875		234	1.013	KIGGAN_20925	full_length_bakta_CDS
NC_003143	4506335	4507449	1115	bcsZ		900	0.807	bcsZ	full_length_bakta_CDS
NC_003143	4510533	4511551	1019	dppB		948	0.93	dppB	full_length_bakta_CDS
NC_003143	4521999	4522793	795	YPO_RS21170		801	1.008	KIGGAN_21230	full_length_bakta_CDS
NC_003143	4541389	4541938	550	rhuM		564	1.025	KIGGAN_21305	full_length_bakta_CDS
NC_003143	4542896	4543164	269	YPO_RS21260		0	0.0		no_bakta_CDS
NC_003143	4543683	4544560	878	YPO_RS21270		486	0.554	KIGGAN_21325	shorter_bakta_CDS
NC_003143	4569065	4569466	402	YPO_RS21385		465	1.157	KIGGAN_21450	full_length_bakta_CDS
NC_003143	4569557	4571510	1954	selB		1773	0.907	selB	full_length_bakta_CDS
NC_003143	4589783	4589896	114	YPO_RS22595		0	0.0		no_bakta_CDS
NC_003143	4607835	4608450	616	YPO_RS21550		315	0.511	KIGGAN_21635	shorter_bakta_CDS
NC_003143	4627368	4627516	149	YPO_RS22355		0	0.0		no_bakta_CDS
NC_003134	31375	32069	695	YPO_RS00660		681	0.98	spo0J	full_length_bakta_CDS
NC_003134	67731	67851	121	YPO_RS00890		0	0.0		no_bakta_CDS

replicon	ncbi start	ncbi end	ncbi span	ncbi gene	ncbi product	bakta span	bakta frac of ncbi	bakta gene	category
NC_003134	73697	73810	114	YPO_RS22615		0	0.0		no_bakta_CDS
NC_003134	76438	76910	473	YPO_RS00940		471	0.996	ykfC	full_length_bakta_CDS
NC_003134	77728	77994	267	YPO_RS00945		318	1.191	KIGGAN_22785	full_length_bakta_CDS
NC_003134	79468	79742	275	YPO_RS21935		0	0.0	KIGGAN_22795	no_bakta_CDS
NC_003131	1956	2204	249	YPO_RS00060		306	1.229	KIGGAN_21880	full_length_bakta_CDS
NC_003131	2307	2951	645	YPO_RS00065		486	0.753	KIGGAN_21890	shorter_bakta_CDS
NC_003131	5099	5391	293	YPO_RS00090		201	0.686	KIGGAN_21915	shorter_bakta_CDS
NC_003131	9083	10228	1146	YPO_RS00115		0	0.0	insO	no_bakta_CDS
NC_003131	10335	11073	739	traT		618	0.836	traT	full_length_bakta_CDS
NC_003131	11113	11403	291	YPO_RS00125		0	0.0		no_bakta_CDS
NC_003131	14382	14663	282	YPO_RS00145		0	0.0		no_bakta_CDS
NC_003131	15628	15846	219	YPO_RS00155		0	0.0		no_bakta_CDS
NC_003131	15976	16293	318	YPO_RS00165		216	0.679	KIGGAN_21985	shorter_bakta_CDS
NC_003131	18189	18826	638	YPO_RS00180		0	0.0	KIGGAN_21995	no_bakta_CDS
NC_003131	45581	46610	1030	YPO_RS00360		0	0.0	tnp	no_bakta_CDS
NC_003131	46629	46901	273	YPO_RS00365		0	0.0	KIGGAN_22175	no_bakta_CDS
NC_003131	48811	49047	237	YPO_RS00375		0	0.0		no_bakta_CDS
NC_003131	49047	49625	579	YPO_RS00380		477	0.824	KIGGAN_22185	full_length_bakta_CDS
NC_003131	49890	49994	105	YPO_RS21890		0	0.0		no_bakta_CDS
NC_003131	54384	54720	337	YPO_RS00400		195	0.579	KIGGAN_22210	shorter_bakta_CDS
NC_003131	54702	54880	179	tnpB		0	0.0		no_bakta_CDS
NC_003131	57858	58106	249	YPO_RS00425		369	1.482	KIGGAN_22240	full_length_bakta_CDS
NC_003131	58250	58453	204	YPO_RS00430		0	0.0		no_bakta_CDS
NC_003131	59970	60346	377	YPO_RS00450		0	0.0	KIGGAN_22260	no_bakta_CDS
NC_003131	60322	60495	174	YPO_RS22470		195	1.121	KIGGAN_22260	full_length_bakta_CDS
NC_003131	62474	62812	339	YPO_RS00460		381	1.124	KIGGAN_22280	full_length_bakta_CDS
NC_003131	68980	69378	399	YPO_RS00495		432	1.083	dnaC	full_length_bakta_CDS
NC_003131	69378	70304	927	istA		927	1.0	istA	full_length_bakta_CDS

source: results/qc/pseudogene_entanglement.tsv

Table S7. Interpretation-layer guidance for the four settings of Section 4.7.

setting	current practice	the rule this paper gives
Laboratory-derived vs naturally circulating	SNP distance to a reference; a small distance is read as the same as a known strain	SNP distance is invalid for this question; a 102.6 kb deletion contributes zero. Check the deletion profile independently: >=10 kb uncalled runs plus the replicon three-state
Including an archive genome in a phylogeny or PAV analysis	Dereplicate by strain field, or by an ANI or SNP threshold	Dereplication must carry a deletion-profile concordance constraint, and that constraint must distinguish absent from not submitted

setting	current practice	the rule this paper gives
Reporting 'X % of strains in a region lack F1'	Compute a frequency from public genomes	That is an archive frequency: an upper bound, reportable only as an interval and only for closed genomes
Deciding whether an absence is biological	Reference-relative absence is scored as a loss	Determine polarity first; the dataset's most frequent absence is AMBIGUOUS and 2 of 13 recurrent regions are gains of the reference lineage

source: Section 4.7

Table S8. The dereplication analysis repeated against a second reference for the same 256 genomes. Coordinates are not comparable between the two references; read each column within itself.

quantity	CO92 reference	Antiqua reference
Reference accession	GCF_000009065.1	GCF_000013825.1
Biovar	Orientalis	Antiqua
Assembly length (bp)	4,829,855	4,879,836
CheckM completeness / contamination (%)	98.69 / 0.86	97.7 / 0.96
Union variant sites (Tier A)	12,926	12,835
Same-strain pairs merging at t = 15	3 of 4	3 of 4
Same-strain pairs merging at t = 20	4 of 4	4 of 4
Largest cluster at t = 20	12	12
CO92 pair, raw distance	13	13
CO92 pair, distance alone	merged	merged
CO92 pair, conservative constraint	separated	separated
Multi-member clusters, no constraint	36	34
Discordant clusters, resolved	24 (21 / 3)	25 (23 / 2)
Discordant proportion	66.7 %	73.5 %
Genomes retained, off -> conservative	175 -> 209	175 -> 208

source: results/ref2/numbers.json and verify.json; results/qc/variant_freq_strata.tsv